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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 8 Compendium:
Control Systems & Variable Rate Technology (VRT)

From Data to Action: Automation Mechanics

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Contents

1 1. Introduction to Control Systems **2**
1.1 Definition . . . 2

2 2. Control System Architectures **2**
2.1 1. Open Loop Control . . . 2
2.2 2. Closed Loop Control (Feedback Control) . . . 2
2.3 PID Control Logic . . . 2

3 3. Electronics Standardization: ISOBUS **3**
3.1 ISO 11783 (ISOBUS) . . . 3
3.1.1 Key Components of ISOBUS . . . 3

4 4. Variable Rate Technology (VRT) **3**
4.1 Method A: Map-Based VRT . . . 3
4.2 Method B: Sensor-Based VRT (Real-Time) . . . 4

5 5. Mechanics of VRT: Liquids (Sprayers) **4**
5.1 Method 1: Pressure Control (The Old Way) . . . 4
5.2 Method 2: Pulse Width Modulation (PWM) (The Modern Way) . . . 4

6 6. Mechanics of VRT: Granular (Fertilizer/Seed) **4**
6.1 1. Gate Height (Variable Orifice) . . . 4
6.2 2. Belt/Chain Speed (Variable Speed) . . . 5
6.3 3. Electric Drives (Planters) . . . 5

7 7. Automatic Section Control (ASC) **5**
7.1 Benefits . . . 5

8 8. Lecture Assessment **6**
8.1 Part A: Standard Concept Questions . . . 6
8.2 Part B: Advanced Analysis (Data & Engineering Logic) . . . 6

1 1. Introduction to Control Systems

In the "PA Cycle" (Measure-Analyze-Decide-Act), we have covered measurement and analysis. Lecture 8 focuses on **Act**. How do we mechanically ensure that the decision made by the computer is executed physically in the field?

1.1 Definition

A **Control System** is a device or set of devices that manages, commands, directs, or regulates the behavior of other devices or systems.

The Agricultural Problem

A tractor sprays chemicals at a Target Rate of 200 Liters/Hectare.

- If the tractor slows down (going uphill), the sprayer must *reduce* flow to maintain 200 L/ha.
- If the tractor speeds up (going downhill), the sprayer must *increase* flow.

This dynamic adjustment requires a **Closed-Loop Control System**.

2 2. Control System Architectures

2.1 1. Open Loop Control

This is a "set it and forget it" system. There is no feedback mechanism.

- **Example:** A simple electric fan. You set the dial to "High". If the voltage drops, the fan slows down, but the switch doesn't know and doesn't care.
- **Ag Example:** Old friction-wheel fertilizer spreaders.

2.2 2. Closed Loop Control (Feedback Control)

The system continuously monitors the output and compares it to the target.

- **Sensor:** Measures the actual output (e.g., Flow Meter).
- **Controller:** Compares Actual vs. Target. Calculates Error.
- **Actuator:** Adjusts the machine to reduce Error.
- **Feedback Loop:** The process repeats continuously (e.g., 10 times per second).

2.3 PID Control Logic

Modern agricultural controllers (ECUs) use **PID Logic** to drive smoothly.

- **Proportional (P):** Reaction to current error. "I am far off target, turn hard."

- **Integral (I):** Reaction to past error accumulation. "I have been slightly off for a long time, nudge it back."
- **Derivative (D):** Reaction to future rate of change. "I am approaching the target too fast, slow down to avoid overshooting."

3 3. Electronics Standardization: ISOBUS

Before the year 2000, every tractor brand had different plugs. You could not plug a Case IH planter into a John Deere tractor.

3.1 ISO 11783 (ISOBUS)

ISOBUS is the international standard for agricultural electronics communication. It allows "Plug and Play" compatibility.

3.1.1 Key Components of ISOBUS

1. **Universal Terminal (UT):** One screen in the cab controls any implement. You don't need 5 different monitors.
2. **Task Controller (TC):** The software that handles GPS maps, section control, and data logging.
3. **TECU (Tractor ECU):** The tractor shares data (speed, PTO rpm, hitch position) with the implement.

4 4. Variable Rate Technology (VRT)

VRT is the ability to vary the rate of input application across a field. There are two primary methods of achieving this.

4.1 Method A: Map-Based VRT

A prescription map (Rx Map) is created *before* entering the field.

1. Agronomist analyzes Soil/Yield maps.
2. Creates a Zoned Map (Zone 1 = 100 kg/ha, Zone 2 = 150 kg/ha).
3. Uploads map to Tractor Computer.
4. **GPS** determines position $\rightarrow$ Computer looks up Rate $\rightarrow$ Controller adjusts Valve.

Pros: Allows for human verification and complex agronomic logic. *Cons:* Reactive (based on old data).

4.2 Method B: Sensor-Based VRT (Real-Time)

The measurement and application happen simultaneously.

1. Sensor (e.g., GreenSeeker) scans crop.
2. Algorithm calculates N-requirement instantly.
3. Controller adjusts Valve immediately.

Pros: Highly responsive to current conditions. No map needed. *Cons:* Limited to what the sensor can see "right now."

5 5. Mechanics of VRT: Liquids (Sprayers)

How do we physically change the flow rate of a liquid sprayer while driving?

5.1 Method 1: Pressure Control (The Old Way)

- **Mechanism:** A butterfly valve throttles the flow.
- **Problem:** Changing pressure changes **Droplet Size**.
- *Physics:* To double the flow, you must quadruple the pressure. High pressure = Tiny droplets (Drift risk). Low pressure = Huge droplets (Poor coverage).

5.2 Method 2: Pulse Width Modulation (PWM) (The Modern Way)

This is the standard for high-precision spraying (e.g., Capstan, Raven Hawkeye).

- **Mechanism:** An electric solenoid on *every nozzle* turns ON and OFF very fast (10Hz or 10 times/second).
- **Control:** We change the **Duty Cycle** (the % of time the nozzle is open).
- **Benefit:** Pressure stays constant (consistent droplet size), but Flow Rate changes linearly.

$$DutyCycle(\%) = \frac{Time_{on}}{Time_{total}} \times 100 \tag{1}$$

6 6. Mechanics of VRT: Granular (Fertilizer/Seed)

How do we vary the rate of solid materials?

6.1 1. Gate Height (Variable Orifice)

An electric linear actuator raises or lowers the rear gate of the spreader.

- *Analogy:* Opening a door wider to let more people through.

6.2 2. Belt/Chain Speed (Variable Speed)

A hydraulic motor drives the conveyor belt feeding the spinners.

- *Mechanism:* The hydraulic valve opens/closes to speed up/slow down the motor.
- *Analogy:* A treadmill running faster.

6.3 3. Electric Drives (Planters)

Modern precision planters replace the ground-drive chain with small electric motors on **every seed meter**.

- *Precision:* Can control population row-by-row. Can stop instantly (Section Control).

7 7. Automatic Section Control (ASC)

ASC uses GPS to automatically shut off sections of the implement (sprayer boom sections or planter rows) when they pass over an area that has already been treated or is outside the field boundary.

7.1 Benefits

1. **Cost Savings:** Reduces input overlap by 5–15%, especially in irregular shaped fields (common in Tanzania).
2. **Crop Safety:** Prevents "double dosing" which can burn the crop (chemical burn) or cause lodging (too much seed/fertilizer).
3. **Operator Fatigue:** The driver doesn't have to manually flip switches at the end of every row.

8 8. Lecture Assessment

8.1 Part A: Standard Concept Questions

1. Define a **Closed-Loop Control System**. Why is it necessary for agricultural machinery?
2. What is **ISOBUS** (ISO 11783), and what "headache" did it solve for farmers?
3. Explain the function of the **Universal Terminal (UT)** in a tractor cab.
4. Contrast **Map-Based VRT** and **Sensor-Based VRT**. Give one example scenario for each.
5. Why is controlling sprayer flow rate via **Pressure** problematic for droplet size?
6. Explain **Pulse Width Modulation (PWM)** in the context of spray nozzles. What stays constant, and what changes?
7. How does a **Variable Rate Seeder** typically adjust the seed population? (Mechanism).
8. Define **Automatic Section Control**. What are its two main economic benefits?
9. What is the role of an **Actuator** in a control system? Give an example of a hydraulic actuator.
10. Explain the function of the **Task Controller (TC)** in an ISOBUS system.

8.2 Part B: Advanced Analysis (Data & Engineering Logic)

1. **PWM Duty Cycle Calculation:** A sprayer nozzle operates at 10 Hz (10 cycles per second). The target flow rate requires the nozzle to be open for 0.07 seconds per cycle. *Task:* Calculate the **Duty Cycle (%)**. If the max flow of the nozzle is 2.0 Liters/min, what is the effective flow rate at this duty cycle?
2. **VRT Logic Algorithm:** You are programming a Variable Rate Fertilizer Spreader for a sugar estate. *Task:* Write a pseudocode logic block using the following inputs:
   - GPS Speed (V)
   - Swath Width (W)
   - Target Rate from Map (R)
   - Calibration Factor (K)

   Output: Hydraulic Motor RPM. *Hint:* Mass Flow = Rate $\times$ Speed $\times$ Width.
3. **ROI of Section Control:** A farmer in Mbeya grows wheat on a 500-hectare irregular field. Due to triangular shapes, his manual overlap is 12%. *Task:* If he installs Automatic Section Control, overlap drops to 2%. Seed costs TZS 100,000 per hectare. Fertilizer costs TZS 200,000 per hectare. Calculate the total savings in TZS for one season.
4. **ISOBUS Compatibility Check:** Visit the **AEF (Agricultural Electronics Foundation)** Database (or search their website concepts). *Task:* Explain the concept of a "Functionality Check." If a Tractor supports UT (Universal Terminal) but the Implement requires TC-GEO (Geo-based Task Control), will Variable Rate Mapping work? Explain why or why not.

5. **System Design - Retrofit:** A smallholder cooperative has an old mechanical boom sprayer (PTO pump, manual valves). *Task:* Design a "Retrofit Kit" to convert this into a Rate Controller system. List the specific electronic components you would need to buy (Sensors, Actuators, Console). Draw a simple block diagram of how they connect.